



Depression detection in read and spontaneous speech: A Multimodal approach for lesser-resourced languages

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ARTICLE INFO

Keywords:

Affective computing
BERT embeddings
Computational paralinguistics
Depression detection
Multimodal fusion
Neural networks

ABSTRACT

The global prevalence of depression highlights the need for innovative early detection methods. Technological advances have driven novel speech analysis strategies for depression identification, yet existing methods often overlook the nuances between read and spontaneous speech and the variability in real-world data. This research introduces a multimodal approach, employing a hybrid model that combines Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN) for detailed audio analysis and a pre-trained BERT model for textual insights. Using a comprehensive corpus of 228 recordings, which includes 64 cases of depression clinically diagnosed by psychiatrists, this study refines a speech-based detection technique reflective of real-world scenarios. Through decision-level fusion, this methodology achieves an accuracy rate of 94.30% and an F-Score of 94.51%, outperforming existing benchmarks in the detection of depression in both read and spontaneous speech. An in-depth feature correlation analysis reveals that spontaneous speech in depressed individuals exhibits pronounced spectral patterns, particularly in Mel-frequency cepstral coefficients (MFCC) interrelationships, whereas read speech retains a subtle but significant diagnostic value. These findings not only show the effectiveness of the multimodal approach but also highlight its better diagnostic precision in the identification of depression, thus setting a new standard for depression diagnosis. By emphasising the critical role of multimodal data analysis, this research significantly advances mental health diagnostics, offering valuable insights for medical practitioners, scholars, and technologists, and represents a considerable leap forward in achieving more accurate and accessible mental health diagnostics. Our code is publicly available at <https://github.com/56kd/MultimodalDepressionDetection>.

1. Introduction

The World Health Organisation recently reported that depression impacts 3.8% of the global population [1]. Despite the prevalence, 63.6% of these cases remain undiagnosed, partly attributed to the limited availability of healthcare services and, in part, to the financial constraints that prevent many individuals from accessing necessary medical care [2]. In response to this challenge, the computing community has significantly advanced the development of technologies aimed at detecting depression using traditional unimodal approaches that use audio, text, or video data [3–7]. While previous multimodal approaches have explored various combinations of audio, text and video data for depression detection [8–10], the distinctive aspect of our approach lies in its hybrid nature, integrating both acoustic features of speech and its semantic content. Unlike methods solely reliant on one modality, our approach capitalises on the strengths of each data type – audio and text – synergistically, offering a more comprehensive and nuanced analysis of depression indicators.

Because multimodal frameworks can significantly increase model performance when working with data that contains many modalities, like audio, text, and video [11–13], they have been used by researchers for speech-based depression diagnosis. However, the lack of medically verified data sources in the models, overlooking the interplay between different speech modalities, and the lack of evaluation in low-resource scenarios, such as non-English settings affected by limited data size, as well as reliance on automated transcription generation without verification, affect the reliability and generalisability of the studies. A unique aspect of this research is the utilisation of *The Androids Corpus* [6], a dataset comprised of both read and spontaneous speech samples in Italian. This dataset enables the first-time application of multimodal depression detection techniques – integrating data from speech acoustics and linguistic content – to an Italian language setting. The corpus is characterised by its clinically validated distinction between depressed and non-depressed speakers, made by professional psychiatrists, thereby focusing on depression detection based on clinical

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<https://doi.org/10.1016/j.bspc.2025.107959>

Received 28 May 2024; Received in revised form 24 February 2025; Accepted 14 April 2025

Available online 27 April 2025

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evaluations rather than through self-assessment questionnaires which are known to be affected by various biases [14].

By analysing feature correlations between read and spontaneous speech, the study reveals unique linguistic and paralinguistic attributes indicative of depressive states, underscoring the importance of a multimodal data examination in comprehending mental health complexities. The proposed research extends beyond demonstrating the multimodal model's superior performance compared to unimodal approaches. The study investigates the acoustic and linguistic features most indicative of depression, thus ensuring the development of targeted and effective diagnostic tools. This approach advances the fields of mental health diagnostics and computational linguistics, offering new insights into the complex landscape of depression detection. This research makes the following contributions:

- A multimodal approach based on language and paralinguistics is used to detect depression in clinically verified patients through *read*, *spontaneous*, and *transcribed* speech. This research harnesses the full spectrum of vocal expression as a diagnostic tool, overcoming the limitations of previous research that focused on narrower data scopes.
- Recognising the dominance of English in available research corpora, we have opted for an Italian dataset, which is not as commonly represented in publicly available datasets. The generalisability of the state-of-the-art solutions is limited by the rigidity of the English public datasets, especially in non-English settings with limited data [6,15,16]. The choice of an Italian dataset not only tests the transferability of techniques developed for one language to another – verified by the effectiveness of the Italian BERT [17] – but also allows us to explore the cultural impact of the proposed approach. In particular, Italian exhibits unique lexical, syntactic, and idiomatic nuances shaped by its cultural and social context [18]; assessing performance in this environment highlights how cultural references or specific linguistic patterns may affect both data interpretation and model outcomes, thereby offering deeper insights into the model's adaptability and potential biases.
- An investigation of the relationship between depression and speech type, employing *read* and *spontaneous* speech samples. This exploration is crucial for understanding how different modes of speech reflect underlying depressive states.
- By demonstrating an effective approach that outperforms state-of-the-art baselines in detecting depression using speech with clinically verified data, our research establishes new benchmarks in the field.

The rest of the paper is structured as follows. Section 2 presents a comprehensive review of state-of-the-art methodologies for depression detection utilising multimodal data sources and analysis techniques. Section 3 highlights the development phases for multimodal depression detection using deep learning, the dataset collection and preparation process, the pipeline of the deep learning algorithms investigated in the study, audio and text feature extraction techniques, feature correlation between audio features, multimodal fusion techniques and rationale for the choice of methodology. Section 4 describes the research questions, experiments, results and discussion to validate the contribution of the proposed approach. Section 5 outlines the research limitations. Finally, Section 6 presents the summary of the research work and future directions.

2. Literature review

As shown in Table 1, multimodal approaches have been investigated to improve depression identification. These approaches leverage audio, video, and text data in combination with advanced machine learning techniques. This section examines key studies that explore various aspects of automatic depression detection, such as the modalities, models, data sources, extracted features, and fusion method.

2.1. Merits of previous approaches

- Use of Diverse Data Sets and Multimodal Data Sources: The incorporation of various data types such as audio, video, and text data [8–10] enhances the depth of analysis and understanding of depression.
- Advanced Neural Network Architectures: Techniques like LSTM and CNN [19–21] are used for sophisticated feature extraction and classification, improving the sensitivity of detection models. Intricate features extracted include MFCCs [22] and BERT embeddings [9].
- Variation in Fusion Types: Employing different fusion methods such as feature-level [9,19,20,22–25] and decision-level [8,10,21,26] fusion allows for a nuanced integration of data modalities, which can leverage the strengths of each type of data more effectively.

2.2. Weaknesses of previous approaches

- Lack of Medically Verified Data Sources in Most Models: Excluding paper [26], none of the studies referenced in Table 1 use a medically verified corpus, instead relying on self-rated depression scales (SDS) completed by participants [25] or automated interviews and questionnaires [8–10,19,20,24,27]. This affects the reliability and generalisability of the findings.
- Overlooking the Interplay Between Different Speech Modalities: All the studies detailed in Table 1 focus purely on depression detection and, therefore do not explore the nuanced differences between read and spontaneous speech, potentially missing crucial indicators of depression.
- Limited Variation of Language Corpora: Most research focuses on English, with limited exploration of other languages. Studies [26] (Italian) and [22] (Chinese) are exceptions, but they highlight the general scarcity of non-English data in the field, restricting the understanding of depression's cultural and linguistic impacts.
- Reliance on Automated Transcription Generation: All the studies in Table 1, excluding [25,26], rely on speech-to-text software to generate text data, which may introduce errors that affect the validity of findings. Misrecognition and omissions can skew data analysis, especially in research demanding precise language. Variations in speech patterns, accents, and dialects can further degrade data reliability.

2.3. Proposed approach to bridge research gaps

- Utilisation of Clinically Verified Data Sources: Ensuring the data used in the study is medically verified adds a robust foundation to the detection methodology, enhancing its credibility and reliability.
- Exploration of the Interplay Between Read and Spontaneous Speech: The research specifically focuses on this aspect, aiming to understand how different speech modalities reflect underlying depressive states, which has been neglected in previous studies.
- Utilisation of an Italian Corpus: The research is conducted using an Italian corpus, addressing the gap in linguistic diversity by moving beyond the predominantly used English datasets. This approach allows for exploring cultural and linguistic impacts on depression detection that have been less studied.

Table 1
Summary of related works on speech-based multimodal depression detection.

Study	Modalities used	Speech-To-Text used?	PModels used	Extracted features	Fusion method	Medically verified data source?
[19]	Audio, Text	Yes	Audio: CNN; Text: LSTM	Audio: Normalised Amplitude Quotient, Open Quotient, Harmonic Difference, Maximum Declination Rate, Peak Slope; Text: Sentiment analysis	Feature-Level	No
[20]	Audio, Text	Yes	Audio: CNN; Text: BERT-CNN, BiLSTM	Audio: Log-mel spectrogram; Text: BERT embeddings	Feature-Level	No
[21]	Audio, Text	Yes	Audio: CNN; Text: LSTM, TCN	Audio: MFCCs, ZCR, RMS energy, Voicing Probability, F0; Text: Word embeddings	Decision-Level	No
[8]	Audio, Text Video	Yes	Audio: LSTM; Text: Bi-LSTM with Attention; Video: LSTM	Audio: Pitch, tone, timbre Text: Sentence embeddings Video: 3D facial landmarks, eye gaze, head pose, facial action unit	Decision-Level	No
[9]	Audio, Text Video	Yes	Audio: GCNN-LSTM; Text: BERT-CNN; Video: GCNN-LSTM	Audio: Deep spectrum features; Text: BERT embeddings; Video: Deep visual representations	Feature-Level	No
[10]	Audio, Text Video	Yes	Audio: LSTM; Text: LSTM; Video: Feed-forward network	Audio: MFCCs, pitch tracking, glottal source parameters; Text: Word embeddings; Video: Facial landmarks, head position, gaze tracking, facial action units	Decision-Level	No
[24]	Audio, Text	Yes	Audio: LSTM; Text: LSTM	Audio: MFCCs, pitch tracking, glottal source parameters; Text: Word embeddings	Feature-Level	No
[25]	Audio, Text	Not specified	Audio: GRU; Text: BiLSTM with Attention	Audio: Mel spectrograms; Text: Sentence embeddings	Feature-Level	No
[26]	Audio, Text	No	Audio: Bi-LSTM; Text: Bi-LSTM	Audio: MFCCs, pitch tracking; Text: Word embeddings	Decision-Level	Yes
[27]	Audio, Text	Yes	Audio: 1D CNN; Text: BiLSTM with Attention	Audio: Mel spectrograms; Text: Word embeddings	Feature-Level	No
[23]	Audio, Text	Yes	Audio: CNN; Text: CNN, Bi-LSTM	Audio: Spectrograms, MFCCs, Pitch, Tone, Rhythm, Stress, Voice quality, Articulation, Intonation; Text: Word embeddings, Sentiment analysis, Linguistic metadata	Feature-Level	No
[22]	Audio, Text	No	Audio: SVM; Text: Bi-LSTM, SVM	Audio: fundamental frequency, MFCCs, zero-crossing rate; Text: Semantic features	Feature-Level	No

- **Enhanced Transcription Accuracy:** Unlike previous studies that depend on a single speech-to-text service [8–10,19–21,23,24,27], our approach uses multiple services with manual, side-by-side comparisons to resolve discrepancies and adopt a majority consensus, thereby minimising errors and enhancing the reliability of transcriptions for diagnostic purposes.
- **Comprehensive Multimodal Analysis:** By integrating both linguistic and paralinguistic features and employing multimodal data (audio, text), the approach aims to capture a complete behavioural signature of depression, thereby addressing the limitations of unimodal and less comprehensive methods.

3. Methodology

The framework for the machine learning-based approach for depression detection from speech is shown in Fig. 1. This framework is divided into four development phases which encompass — the collection and

processing of speech datasets (Section 3.1), the extraction of audio and text features and their subsequent correlation analysis (Section 3.2), the development and evaluation of machine learning models (Section 3.3), the integration of multimodal fusion techniques and the evaluation of fusion strategies (Section 3.4). The individual phases of the proposed framework for multimodal depression detection, utilising both read and spontaneous speech, are detailed below.

3.1. Dataset collection and processing

The dataset is sourced from the Androids Corpus [6], a publicly available resource containing 228 audio recordings in .wav format. These recordings were collected from 118 native Italian speakers, including both individuals diagnosed with depression and healthy control subjects, making them suitable for comparative analysis. Each recording includes either interview-style responses or task-based reading segments, allowing for a comprehensive examination of the acoustic and linguistic markers associated with depression. The distribution of

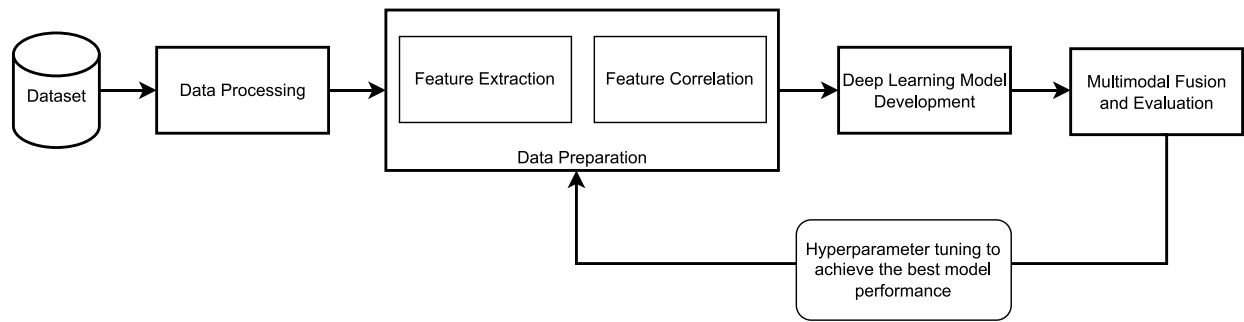


Fig. 1. Proposed multimodal depression detection framework using read and spontaneous speech.

Table 2
Participant distribution across Reading Task (RT) and Interview Task (IT).

Group	RT only	IT only	RT and IT	Total
Control	2	0	52	54
Depression	0	6	58	64
Total	2	6	110	118

the dataset is detailed in Table 2. In this section, we will describe the process of preparing the dataset (Section 3.1.1), focusing on the handling of audio data and the generation of transcriptions using audio-to-text tools (Section 3.1.2).

3.1.1. Audio data preparation

The audio data was sourced from two distinct tasks: interviews and a reading task. The data set was processed specifically in both contexts to focus on the contributions of the patients. In the interview task, spontaneous speech samples were collected as participants responded to personal and general questions, while participants read a short story in the reading tasks.

3.1.2. Transcription generation methodology

Transcriptions of the audio files were necessary to complement the audio data with textual information. Given the importance of accuracy in medical datasets, particularly for training diagnostic models, multiple speech-to-text services were used to generate these transcriptions, such as Buzz, powered by OpenAI Whisper,¹ Google’s Cloud Speech API² and Descript.³ Each service processed the same set of audio files, and their outputs were manually compared. The review process involved identifying transcription errors, misinterpretations, and omissions. Manual verification of the transcriptions was done by carefully analysing each discrepancy through a side-by-side comparison of the outputs from the speech-to-text services. The service outputs were evaluated for how they handled differences. Buzz was determined to produce the most accurate transcriptions. This decision was based on the service’s ability to align with at least one other service in cases where outputs differed, effectively using a majority consensus approach to validate the accuracy of the transcriptions. This comparison method highlighted Buzz’s reliability, as it consistently matched the transcription of at least one other service, ensuring a high level of agreement and making it the preferred choice for accuracy. Similar studies have shown that OpenAI’s transcription model, Whisper automatic speech recognition (ASR) [28], which powers Buzz used in this research, consistently demonstrated superior accuracy when compared to Google’s,

Azure’s, Descript, and Amazon’s speech-to-text software.^{4,5,6,7,8} This further justifies the selection of Buzz in the transcription process for the data used in this research.

3.2. Data preparation

During this phase, we extract features from audio and textual data that indicate depressive states. These features are essential for conducting feature correlation analysis and inputting into deep learning models. These models help identify patterns and distinctions associated with depressive speech. The feature extraction and correlation phase methodology are described in detail in the following.

3.2.1. Feature extraction

This phase involves extracting features from the audio and textual data from the automated speech-to-text transcription. The audio and textual feature extraction process is described below.

Audio Feature Extraction: Feature extraction from audio data was performed using openSMILE [29], widely recognised as the industry standard for feature extraction in audio signal processing. It provides robust and reliable tools essential for advanced speech and emotion analysis [6,7,21,22]. A standard set of features extracted from the audio recordings are described below.

- Root Mean Square (RMS) Energy: Indicates the energy level of the speech, often correlating with the vocal intensity, which can vary significantly in depressive speech.
- Mel-Frequency Cepstral Coefficients (MFCCs): These 12 coefficients represent the short-term power spectrum of sound, capturing key phonetic characteristics essential for recognising changes in speech due to depression.
- Fundamental Frequency (F0): Related to the pitch of the voice, changes in F0 can reflect emotional states and are often lower in depressed individuals.
- Zero-Crossing Rate (ZCR): A measure of the frequency of sign changes in the speech signal, related to the tone of the voice.

¹ <https://github.com/chidiwilliams/buzz>.
² <https://cloud.google.com/speech-to-text/>.
³ <https://www.descript.com/tools/speech-to-text>.

⁴ <https://sstoitsev.medium.com/google-vs-azure-a-speech-to-text-battle-f740aa481e8e>.
⁵ <https://www.gladia.io/blog/openai-whisper-vs-google-speech-to-text-vs-amazon-transcribe>.
⁶ <https://www.captions.ai/blog-post/openai-beats-google>.
⁷ <https://slator.com/how-big-a-deal-is-whisper-for-asr-multilingual-transcription/>.
⁸ <https://slator.com/how-big-a-deal-is-whisper-for-asr-multilingual-transcription/>.

- **Voicing Probability:** Indicates the likelihood of a speech segment being voiced versus unvoiced, reflecting variations in speech dynamics that might be influenced by depression.

This feature set was augmented with regression coefficients, resulting in a total dimensionality of $D = 32$. This feature set was originally developed for the Interspeech Emotion Recognition Challenge in 2009 [30] and has since been widely applied across various domains, particularly for extracting social and psychological insights from speech [6].

Textual Feature Extraction: For the textual component of the dataset, TF-IDF vectorisation [31,32] and BERT model embeddings [33] were employed to transform raw text data into formats suitable for machine learning models:

- **TF-IDF Vectorisation:** This method is effective for the Italian language due to its straightforward approach that does not require language-specific resources beyond a simple tokeniser, which is widely available [34].
- **Bidirectional Encoder Representations from Transformers (BERT) Model Embeddings:** The use of BERT, particularly multilingual models that include Italian, provides a deep linguistic and contextual understanding necessary to detect subtleties in language indicative of depression [17]. The three pre-trained BERT models used to extract embeddings are as follows: *dbmdz/bert-base-italian-xxl-cased*, *dbmdz/bert-base-italian-xxl-uncased*⁹ and *neuraly/bert-base-italian-cased-sentiment*¹⁰ hosted by Hugging Face [35]. The pre-trained Italian BERT models were used to calculate the sentiment scores of tweets and users during the 2020 Italian Constitution referendum [36]; to extract legal holdings from Italian case documents [37]; to recognise dialogue acts and slots in Italian complex dialogues [38] and to create a high-performing clinical information retrieval system from Italian clinical notes [15]. These use cases demonstrate their suitability for Italian speech and text analysis for the detection of depression.

The datasets for text analysis were prepared separately using TF-IDF vectorisation and BERT embeddings. This approach allows flexibility in model training and testing, allowing a comparative study of how each text representation technique performs in the context of depression detection.

3.2.2. Feature correlation

Although the use of feature correlation matrices in automatic depression detection has been explored [7], this study uniquely focuses on comparing these matrices between read and spontaneous speech. The primary objective here is to uncover how the interplay between these two speech modalities impacts the detection of depression, advancing our understanding beyond the existing literature that predominantly concentrates on *read* speech alone.

The methodology focuses on a structured approach to analysing audio data that covers data segmentation, correlation computation and visualisation as described below.

- **Data Segmentation:** Data was segmented based on the combinations of depression status and speech type. This segmentation created four distinct groups: *read* depressed, *read* non-depressed, *spontaneous* depressed, and *spontaneous* non-depressed. This detailed segmentation enables targeted analysis of feature correlations within each specific group, shedding light on the subtle differences and similarities in how depression manifests in read versus spontaneous speech.

- **Correlation Computation and Visualisation:** Correlation matrices were computed for each subset to determine the relationships among the features. These matrices help highlight significant correlations that might distinguish depressive speech from non-depressive speech. Heatmaps were generated to visually represent these correlations using *seaborn*¹¹ and *matplotlib*¹² tools. Heatmaps provide an intuitive understanding of complex data, making it easier to identify and interpret feature interdependencies visually.

The decision to prioritise audio data over text data is based on the reliability of direct measurement from audio signals compared to the potential inaccuracies in automated text transcriptions. Because a native Italian speaker did not manually verify the text data generated from automated services, it might contain errors that could skew the analysis. On the other hand, audio data offers raw, unfiltered insights into speech dynamics, such as tone, pitch, and rhythm, and provides a clearer insight into the vocal attributes crucial for depression detection, such as intonation, stress, and speech rate.

3.3. Development of deep learning models

The proposed approach in Fig. 1 employs deep learning algorithms instead of traditional machine learning models, as deep learning has shown better performance in complex pattern recognition tasks, such as depression detection in speech [6,23]. The deep learning algorithms outlined in Sections 3.3.1 and 3.3.2 were developed for this study and chosen for their proven effectiveness in similar tasks of automated depression detection using audio data, text data, or a combination of both, as shown in Table 1. All models developed in this article were created using the PyTorch framework [39] and employed the *Adam* optimiser [40]. The choice and methodology of the algorithms implemented for the audio and text tasks are described below.

3.3.1. Audio models

In the development and evaluation of all audio models designed to detect depression, a series of five standardised techniques, as defined by the dataset authors were consistently applied to ensure accuracy, robustness, and applicability [6]. The authors' protocol, as described in the dataset, was adhered to for feature extraction and vectorisation, aggregation through majority voting, k-fold cross-validation (including a predefined test-train split) and person-independent experiments. The fifth standardised technique implemented was data augmentation with random noise during training. This augmentation simulates real-world conditions, such as background noise, that could affect audio recordings. By training the models to recognise depression indicators amidst this noise, they become more effective and resilient in diverse acoustic environments. This step is crucial for developing models that maintain high performance regardless of external audio disturbances. The specific hyperparameters for the audio models are detailed in Table 3.

The choice of the audio models investigated in this research is described below.

- **Convolutional Neural Networks (CNN):** CNNs were chosen for their proficiency in feature extraction, which is vital for analysing subtle speech aspects in depression diagnosis. Their capability to autonomously learn from large data volumes positions them as ideal for detecting complex patterns in depressive speech, ensuring comprehensive spectral feature analysis [19–21]. There are also many advanced popular CNN architectures with seminal backbone networks such as ResNet [41] and DenseNet [42], that have significantly improved performance on state-of-the-art

⁹ <https://huggingface.co/dbmdz/bert-base-italian-cased>.

¹⁰ <https://huggingface.co/neuraly/bert-base-italian-cased-sentiment>.

¹¹ <https://seaborn.pydata.org/generated/seaborn.heatmap.html>.

¹² https://matplotlib.org/stable/gallery/images_contours_and_fields/image_annotated_heatmap.html.

Table 3
PHyperparameter configurations for the audio models.

Model	Component	Hyperparameter	Value
1D CNN	Convolutional layers	Kernel size	5
	Pooling layers	Type	MaxPooling1D
	Dropout layers	Dropout rate	0.5
	Fully connected layers	Layer type	Dense (Flatten)
	Training	Epochs	100
PDenseNet CNN	Convolutional layers	Growth Rate	12
	Dense blocks	Block configuration	(6, 12, 24)
	Transition layers	Compression factor	0.5
	Pooling layers	Type	AvgPooling1D
	Fully connected layers	Layer type	Dense (Flatten)
PResNet CNN	Convolutional layers	Kernel size	7
	Convolutional layers	Initial filters	64
	Residual blocks	Blocks per layer	[2, 2, 2]
	Pooling layers	Type	AvgPooling1D
	Fully connected layers	Output units	1
LSTM	Convolutional layers	Epochs	30
	Model configuration	Number of layers	3
	Dropout layers	Dropout rate	0.3
	Training	Epochs	100
Bi-LSTM	Model configuration	Number of layers	3
	Layers and Dropout	Dropout rate	0.3
	Training details	Epochs	100
GRU	Model configuration	Number of layers	1
	Layers and Mechanism	Hidden dimensions	128
	Training details	Epochs	100
Transformer	Positional encoding	Encoding method	Sine and Cosine functions
	Multi-head attention	Focus mechanism	Multi-head attention
	Feedforward neural networks	Layers	Within each Transformer block
	Optimisation details	Epochs	100
Hybrid CNN-LSTM	Convolutional layer	Filters	64
	Pooling layer	Kernel size	5
	LSTM layer	Type	Dimensionality reduction
	Training details	Hidden units	128
	Attention mechanism	Epochs	10 for RT, 15 for IT
Hybrid CNN-BiLSTM	Convolutional layer	Learning rate scheduler	Reduces rate every 30 epochs
	Pooling layer	Integration point	Post-LSTM
	LSTM layer	Purpose	Weigh sequence segments
	Training details		
	Attention mechanism		

benchmark tasks. These sophisticated networks are capable of extracting deeper feature information [41–43]. To further enhance model performance and explore different architectures, standard 1D CNN [44], DenseNet-based CNN, and ResNet-based CNN models were developed. Each model leveraged unique structural strengths: the standard 1D CNN for its simplicity and efficiency, DenseNet for its dense connectivity promoting feature reuse, and ResNet for its residual connections mitigating vanishing gradients. These architectures were tailored to handle the nuances of audio feature data, optimising their ability to capture critical patterns relevant to depression diagnosis. Implementing simple and advanced deep convolutional neural network (CNN) architectures enables us to conduct a comparative analysis of various CNN models. This approach allows us to assess how deeper models influence the detection of depression in both read and spontaneous speech.

Similarly, for the transcribed text, we built CNN models to capture complex patterns in sentence structures, which are crucial for discerning subtle linguistic cues indicative of depression. The

effectiveness of CNNs is significantly enhanced when paired with input embeddings such as TF-IDF and BERT. TF-IDF embeddings assist CNNs by emphasising the most relevant words within texts, thus allowing the network to focus on key terms that might indicate depressive sentiment. Meanwhile, BERT embeddings provide rich, context-aware representations that capture the nuances of language used by individuals, which is critical for understanding the deeper, latent meanings that might signal depression [20]. Together, these embeddings provide a robust framework that enables CNNs to perform with high precision and sensitivity in detecting depressive states from textual data.

- **Long Short-Term Memory (LSTM):** LSTM models are investigated for analysing segmented audio features for detecting depression due to their effectiveness in handling time-series data like audio. Their ability to remember information for long durations is crucial for analysing speech where contextual and temporal dynamics provide significant insights into the mental state of individuals. By modelling both the current input features and the features' history effectively, LSTMs offer a robust framework

for detecting nuanced changes in speech patterns associated with depression [8,10,24].

In addition to applying LSTM models for audio data, we extended our methodology to the textual domain by employing LSTM networks on the transcribed text. LSTMs are suitable for analysing the flow and context of language in the user-generated text, which often contains clues about the user's mental state [10,21]. When combined with BERT embeddings, LSTMs become even more effective. BERT provides deep contextual representations that capture the nuances of language, allowing the LSTM to interpret the subtle meanings embedded in text sequences that may indicate depression. When paired with TF-IDF vectors, LSTMs gain an enhanced ability to discern patterns indicative of depression in textual data.

- **Bidirectional Long Short-Term Memory (Bi-LSTM):** Bi-LSTMs are highly effective for audio data analysis, especially in applications like depression detection from speech, where understanding the full context of temporal sequences is crucial. The choice of Bi-LSTMs is justified by their superior ability to process sequential data that requires insights from both preceding and succeeding elements in the dataset. This dual-direction processing is particularly beneficial for speech analysis, where the context before and after a given moment can provide significant insights into the speaker's emotional state [26].
- **Gated Recurrent Units (GRU):** This research utilises GRUs to analyse audio features for detecting depression, leveraging their efficient handling of dependencies in time-series data, which is essential for capturing the dynamics in speech patterns linked to depressive states. They provide a balanced approach to learning dependencies, making them particularly suitable for audio data where temporal relationships significantly influence detection accuracy [25].
- **Transformers:** The Transformer model, known for its self-attention mechanism [45], is adapted for audio feature analysis to detect depression. Transformers are particularly suited for audio analysis tasks such as depression detection because they can handle long sequences with complex dependencies. The self-attention mechanism allows Transformers to perform deep and nuanced analysis of audio features, crucial for detecting subtle and complex patterns associated with depressive speech [46].
- **Hybrid CNN-LSTM:** The hybrid CNN-LSTM model leverages the strengths of both CNN and LSTM networks, making it highly effective for audio signal analysis where feature extraction and sequential data processing are crucial. This architecture, described in Fig. 2, is utilised to detect depression from audio recordings by extracting robust feature representations with CNNs and analysing temporal dependencies with LSTMs. An attention mechanism is integrated post-LSTM processing to weigh the importance of different segments of the sequence, enhancing the model's focus on more informative parts of the audio that may indicate depression [8]. This model's structure ensures comprehensive learning from both spectral and temporal speech features, making it a powerful tool in the automated detection of depression [47]. The systematic approach to training, including early stopping to prevent overfitting and scheduled learning rate reductions, ensures that the model is both efficient and effective, providing robust predictions that can significantly aid in mental health diagnostics.
- **Hybrid CNN-Bi-LSTM:** This adaptation incorporates bidirectional LSTM layers, which process data from past and future contexts simultaneously, offering a more complete understanding of temporal dependencies. This feature is particularly valuable in tasks like depression detection from audio, where understanding the full emotional context conveyed in speech patterns is crucial [48]. The bidirectional approach allows the model to capture nuances and intonations in audio signals that unidirectional LSTMs might overlook. For example, how the end of a sentence is spoken can

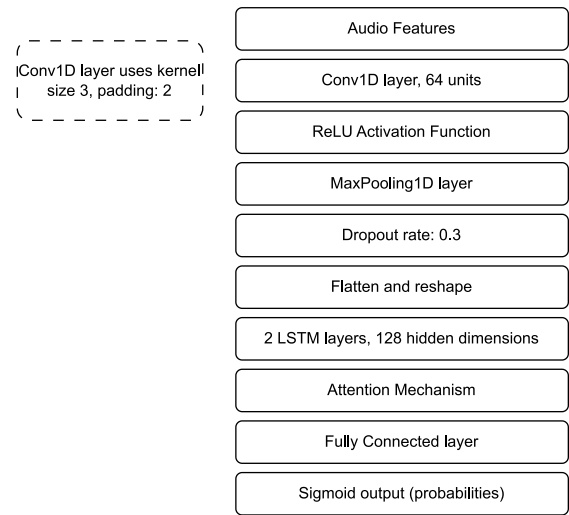


Fig. 2. Architecture of the hybrid CNN-LSTM audio model.

reveal emotional states when considered with the start of that sentence. By analysing these elements from both directions, the CNN-Bi-LSTM can achieve a richer, more accurate detection of depression, potentially improving both the sensitivity and specificity of the diagnostic process [49]. This makes the CNN-Bi-LSTM an important experimental variation to explore, potentially leading to more effective mental health diagnostics through enhanced audio signal analysis.

3.3.2. Text models

The text-based models for detecting depression utilise the same protocols as the audio models. This alignment across modalities ensures that the methodologies are consistent, enhancing the robustness and effectiveness of the text models in diagnosing depression, comparable to the audio models. The hyperparameter configurations of the proposed text classification models are detailed in Table 4. In addition to the CNN and LSTM models described in Section 3.3.1, the BERT model is used to analyse the textual features extracted from transcribed audio data. This model is described below.

- **BERT Models:** BERT's bidirectional training structure allows the model to effectively understand the context from both the left and right sides of a token within the text, making it exceptionally adept at capturing nuanced emotional expressions indicative of mental health states [9]. Only BERT embeddings were used for the BERT model, bypassing the TF-IDF vectors employed in other models. As shown in Fig. 3, majority voting is performed by aggregating model predictions across different text segments (segments are created by grouping every two sentences). Each segment's prediction is considered a vote, and the majority vote determines the final decision for a given text sample. This method enhances decision-making by leveraging the consensus among various text segments.

3.4. Multimodal fusion and evaluation

The existing multimodal representation techniques are classified into two categories — *early* and *late fusion*. Early fusion is usually achieved by concatenating extracted features or raw data at an initial stage, and then one model is trained. Many independent models are trained for each modality when using late fusion, and their outputs are then connected [50,51]. Fusion at the data level is not considered in this research on depression detection as it could lead to the loss

Table 4
PHyperparameter configurations for the text models.

Model	Component	Hyperparameter	Value
Pre-Trained BERT model	Data processing	Max sequence length	512
		Tokeniser padding	True
	Training	Epochs	3
		Weight decay	0.01
	Evaluation	Majority voting threshold	0.5000
CNN with BERT embeddings	Convolutional layer 1	Filters	64
	BatchNorm layer 1	Kernel size	3
		Features	64
		Type	MaxPool1d
	Pooling layer 1	Kernel size	2
		Rate	Rate
	Dropout layer 1	Rate	Rate
	Convolutional layer 2	Filters	32
		Kernel size	3
	BatchNorm layer 2	Features	32
		Type	MaxPool1d
	Pooling layer 2	Kernel size	2
		Rate	0.2
	Dropout layer 2	Rate	0.2
	Fully connected layer	Hidden units	(Input Dimensions/4) * 32
CNN with TF-IDF	Training details	Epochs	100
		Learning rate	0.001
	Loss function	Type	BCEWith-LogitsLoss
	Convolutional layer 1	Filters	64
	BatchNorm layer 1	Kernel size	3
		Num features	64
		Type	MaxPool1d
	Pooling layer 1	Kernel size	2
		Rate	0.2
	Convolutional layer 2	Filters	32
		Kernel size	3
	BatchNorm layer 2	Num features	32
		Type	MaxPool1d
	Pooling layer 2	Kernel size	2
		Rate	0.2
LSTM with BERT embeddings	Dropout layer 2	Rate	0.2
	Fully connected layer	Output units	1
	Training	Epochs	100
		Learning rate	0.001
	Loss function	Type	BCEWith-LogitsLoss
	DataLoader	Batch size	32
	LSTM layer	Hidden dimension	128
	Model configuration	Number of layers	1
		Output units	1
		Epochs	100
	Training	Learning rate	0.001
		Type	BCEWith-LogitsLoss
	Loss function	Type	BCEWith-LogitsLoss
	DataLoader	Batch size	32
	LSTM layer	Hidden dimension	100
LSTM with TF-IDF	Model configuration	Number of layers	1
		Output units	1
		Epochs	100
	Training	Learning rate	0.001
		Type	BCEWith-LogitsLoss
	Loss function	Type	BCEWith-LogitsLoss
	DataLoader	Batch size	32
	LSTM layer	Hidden dimension	100
	Model configuration	Number of layers	1
		Output units	1
		Epochs	100
	Training	Learning rate	0.001
		Type	BCEWith-LogitsLoss
	Loss function	Type	BCEWith-LogitsLoss
	DataLoader	Batch size	32

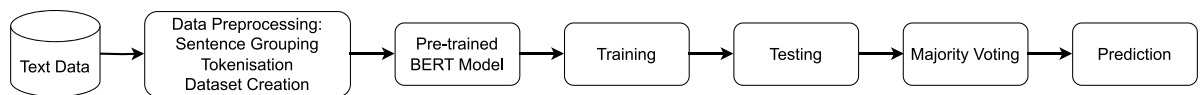


Fig. 3. Architecture of the BERT text model.

of critical modality-specific features. Such features are more effectively captured and utilised when audio and textual data are processed separately [52]. This approach is particularly crucial when analysing read versus spontaneous speech, where distinct characteristics of each speech type must be preserved and accurately assessed for effective depression detection.

In this study, early fusion is performed at the feature level to integrate audio and text data for depression detection, enhancing the

robustness and accuracy of the predictive model. As displayed in Fig. 4, this approach involves the direct concatenation of feature sets from different modalities before input into the predictive model, allowing the model to learn from a more comprehensive feature space that encapsulates audio cues and textual information. In the context of depression detection, combining audio and text features at an early stage allows the model to simultaneously process and learn from both spoken nuances and linguistic content [53]. By applying dimensionality

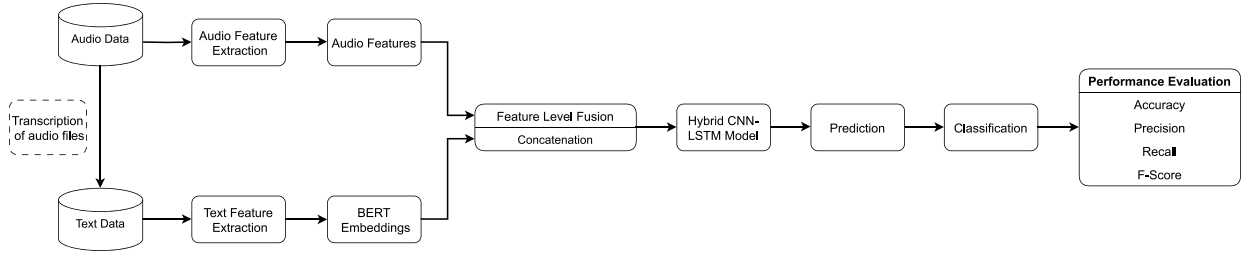


Fig. 4. Feature-level fusion for automated multimodal depression detection.

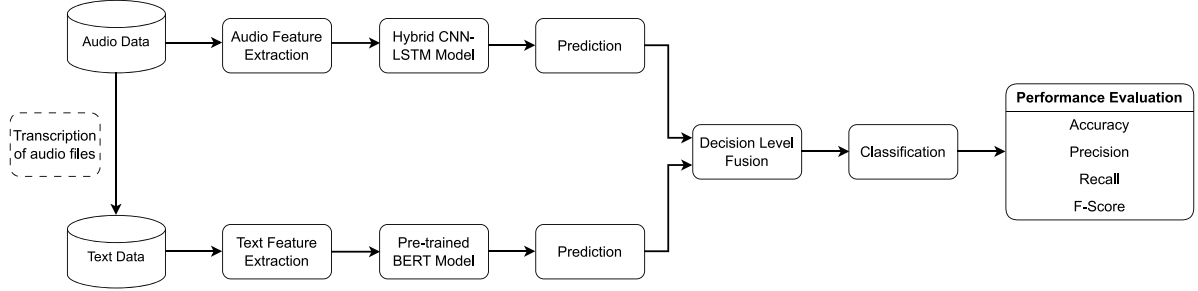


Fig. 5. Decision-level fusion for automated multimodal depression detection.

reduction techniques like Principal Component Analysis [54] after fusion, the model efficiently handles high-dimensional data, focusing on the most informative features across modalities, leading to improved diagnostic performance [55].

Furthermore, late fusion is performed at the decision level, which involves combining the outputs from multiple independent models, each processing different data modalities, to make a final decision, as shown in Fig. 5. This approach is effective in scenarios where specialised models best handle distinct feature sets. Fusing decisions later leverages each model's strengths, potentially leading to more robust and accurate predictions [53]. Decision-level fusion is justified in complex diagnostic tasks like depression detection, where different modalities may provide complementary insights into the patient's condition. For instance, audio models might capture emotional tone and speech patterns, while text models might understand content and linguistic subtleties. Moreover, decision-level fusion enhances the robustness of the diagnosis by aggregating multiple independent judgments, thereby reducing the risk of misdiagnosis due to modality-specific biases or errors.

The decision-level techniques used in this research are as follows.

- **Weighted Average:** Eq. (1) describes the weighted average (mean), where p_i is the prediction score from model i (which could be either an audio or text model), and w_i is the corresponding weight reflecting the model's reliability. In this context, the weights, w_i , are directly based on the accuracy of each model, with higher accuracy resulting in a higher weight for that model. This weighted average approach integrates the strengths of both audio and text-based assessments of depression, allowing for a more nuanced analysis [56]. By blending these modalities, the fusion technique capitalises on the comprehensive information available from different data types, thus enhancing the overall accuracy and robustness of depression detection [57]. The architecture of the weighted average fusion technique is detailed in Fig. 6.

$$\text{Final Score} = \sum_{i=1}^n w_i \times p_i \quad (1)$$

- **Logistic Regression:** The final decision-making process in the provided implementation is achieved through logistic regression, which combines the outputs of individual audio and text models to produce a robust final prediction. The logistic regression

classifier operates as described by Eq. (2).

$$\text{Final Decision} = \sigma \left(\sum_{i=1}^n \beta_i \times p_i + \beta_0 \right) \quad (2)$$

Here, σ represents the logistic function, β_i are the coefficients learned by the regression model and p_i are the outputs from the individual audio and text models. The integration method shown in Fig. 7 does not use the stacking integration technique. Instead, it follows a decision-level fusion approach. While stacking involves training a meta-learner on the outputs of base models using a separate validation set to prevent overfitting [58,59], the method presented here directly combines the probability outputs from the audio and text models through logistic regression.

As shown in Fig. 7, each model's output probability is used as an input feature for the logistic regression classifier. The probabilities from the audio and text models serve as inputs to the summation node, which applies the learned weights w_{audio} and w_{text} to combine the outputs before passing them through a sigmoid function. In this implementation, the logistic regression model learns the optimal combination of these features to maximise predictive performance. By leveraging the strengths of both modalities, this decision-level fusion technique ensures a robust and effective classifier, capable of assessing complex indicators like emotional states from multimodal data. This approach is particularly effective for tasks requiring nuanced classification across modalities [60].

- **Bayesian Approach:** Bayesian methods, which calculate the posterior probability of a decision D given the predictions $p_1, p_2, \dots, p_n$ from various models, utilise prior knowledge about the models' accuracies as shown in Eq. (3).

$$P(D|p_1, p_2, \dots, p_n) = \frac{P(p_1, p_2, \dots, p_n|D) \times P(D)}{P(p_1, p_2, \dots, p_n)} \quad (3)$$

In the context of depression detection, such an approach is instrumental when integrating decision-level outputs from both audio and text models. Here, $P(D)$ is the probability that an individual is depressed, p_1 is the prediction of the audio model and p_2 is the prediction of the text model. As shown in Fig. 8, the prior knowledge is calculated as the percentage of depressed patients in the dataset. By applying Bayesian methods, the integrated system can assess the probability of depression more accurately,

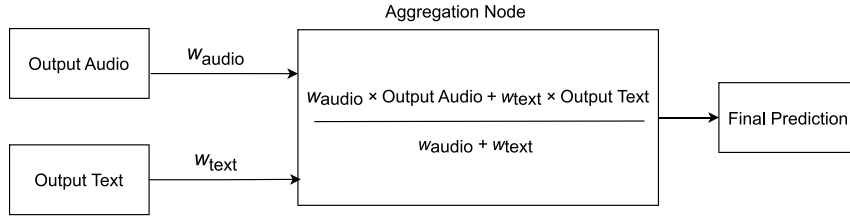


Fig. 6. Weighted average fusion architecture.

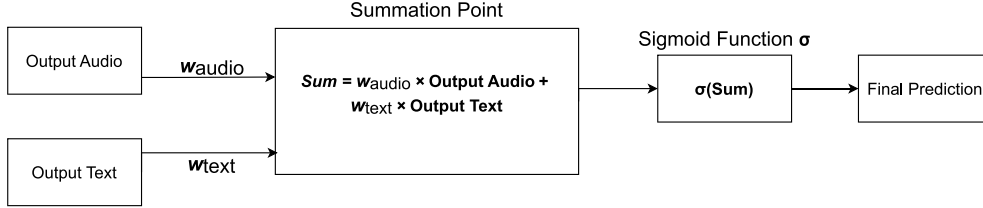


Fig. 7. Logistic regression fusion architecture.

considering each model type's varying reliability and strengths. For instance, audio models might capture nuances in tone and speech patterns, while text models analyse linguistic cues and verbal expressions. Bayesian integration effectively weighs these diverse inputs according to their demonstrated accuracy, thereby enhancing the overall decision-making process in diagnosing depression [61]. This method increases confidence in the results and mitigates the risk of errors due to model-specific biases or anomalies.

- **Gated Multimodal Unit (GMU):** As shown in Fig. 9, the GMU employs a distinct gating mechanism in each of its specialised layers – audio transform, text transform, audio gate, and text gate – to dynamically control the contribution of audio and textual inputs to the final diagnosis. This enables the model to effectively weigh the relevance of each modality in real-time, optimising its predictive accuracy, making it a robust tool for multimodal depression detection in diverse clinical environments [10,26]. This mechanism is implemented following the methodology from [62] as described in Eq. (4), where each modality's data x_{audio} and x_{text} are first transformed via ReLU activated linear layers h_{audio} and h_{text} . Then, gating layers apply a sigmoid (σ) function to compute the gates z_{audio} and z_{text} determining how much each modality contributes to the final output. The final integration h balances the input modalities, optimising predictive accuracy by leveraging both the nuanced tonal information from audio and the semantic richness of the text.

$$\begin{aligned}
 h_{audio} &= \text{ReLU}(W_{audio} \cdot x_{audio}) \\
 h_{text} &= \text{ReLU}(W_{text} \cdot x_{text}) \\
 z_{audio} &= \sigma(W_{gate, audio} \cdot x_{audio}) \\
 z_{text} &= \sigma(W_{gate, text} \cdot x_{text}) \\
 z &= \sigma(W_{gate} \cdot [x_{audio}, x_{text}]) \\
 h &= z_{audio} * h_{audio} + z_{text} * h_{text}
 \end{aligned} \tag{4}$$

Each audio and text model, as well as both methods of multimodal fusion, were evaluated using the metrics of Accuracy, Precision, Recall, and F-score, as detailed in Figs. 4 and 5.

4. Results and analysis

This research focuses on using a multimodal fusion of deep learning models to improve the detection of depression through speech analysis. The study examined two distinct types of speech: *read* and *spontaneous*.

By integrating these modalities, the research utilised the unique characteristics of each speech type to enhance the accuracy and reliability of depression detection. The research questions outlined below were formulated to validate and assess the effectiveness of the multimodal fusion approach.

- RQ1 — What are the differences in depression detection rates between *read* and *spontaneous* speech samples, and which paralinguistic features are most influential in these differences?
- RQ2 — Which machine learning algorithms perform best in detecting depression from speech, and how do they compare to current clinical assessment methods?
- RQ3 — Which method of multimodal fusion demonstrates the highest performance in detecting depression from speech, and how do they compare to current clinical assessment methods?
- RQ4 — How does integrating linguistic and paralinguistic features enhance the accuracy and reliability of depression detection compared to single-mode speech analysis?
- RQ5 — How does the proposed multimodal depression detection technique improve upon current methodologies in affective computing for depression detection?

4.1. RQ1 — What are the differences in depression detection rates between *read* and *spontaneous* speech samples, and which paralinguistic features are most influential in these differences?

To address RQ1, correlation matrices are provided for *read* and *spontaneous* speech from depressed and non-depressed cohorts. This enables a granular analysis of how paralinguistic features differ among these groups. When comparing the matrices, certain features stand out in their association with depression as detailed below.

4.1.1. Read speech depressed vs. Non-depressed

In the Read Depressed matrix displayed in Fig. 10(a), there are conspicuous blocks of strong positive correlations among the MFCCs, particularly in the lower orders ($\text{pcm_fftMag_mfcc}[1-5]$), which suggest these features capture changes in voice quality associated with depression. This pattern diminishes progressively as we move to the higher-order MFCCs. The Zero-Crossing Rate (zcr_sma) and Fundamental Frequency (F0final_sma) features, which are tied to speech rate and pitch, respectively, show less pronounced correlations. Contrastingly, the Read Non-Depressed matrix, depicted in Fig. 10(b) displays a more sparse and weaker correlation pattern across these MFCCs, indicating less uniformity in vocal characteristics. This could imply that while depressed individuals' speech patterns become homogeneous,

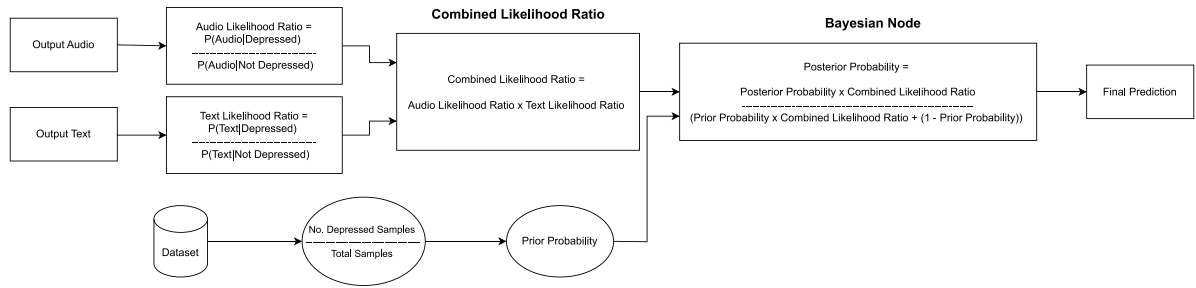


Fig. 8. Bayesian fusion architecture.

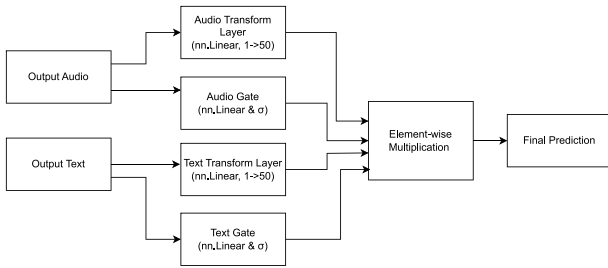


Fig. 9. Gated multimodal unit architecture.

non-depressed individuals maintain varied vocal signatures when reading.

4.1.2. Spontaneous speech depressed vs. Non-depressed

The Spontaneous Depressed matrix shown in Fig. 11(a) presents a more heterogeneous correlation structure, with lower degrees of positive correlations scattered across the MFCCs, suggesting that spontaneous speech elicits a broader array of vocal expressions, even within the depressed cohort. Energy features, such as Root Mean Square Energy (pcm_RMSenergy_sma), do not show the same level of strong correlation seen in read speech, which could reflect the dynamic range of spontaneous emotional expression. In the Spontaneous Non-Depressed matrix, Fig. 11(b), the correlations are significantly lower compared to those in the Spontaneous Depressed matrix, suggesting a greater diversity in vocal expression that is not associated with depressive effects. Features such as Voicing Probability (voicingFinalUnclipped_sma), which relates to the quality of voicing, exhibit a negligible correlation, potentially serving as a discriminant feature in spontaneous speech between depressed and non-depressed individuals.

4.1.3. Interpreting paralinguistic feature influences

When analysing both speech modalities, it becomes evident that depression distinctly influences the correlation patterns among various feature sets. In read speech, the strong interrelationships among the lower-order MFCCs suggest these features capture consistent changes in the vocal characteristics indicative of depression. Conversely, spontaneous speech contrasts with a wider and less defined correlation pattern. Features such as Root Mean Square of Energy and Voicing Probability show a diffuse correlation structure in the Spontaneous Depressed matrix, indicating a diverse expression of vocal features influenced by the individual's affective state. These findings suggest that spontaneous speech captures a richer, more complex range of emotional indicators.

4.1.4. Depression detection implications

The contrast in correlation patterns between *read* and *spontaneous* speech suggests that modality-specific features are essential for effective depression detection. Developing detection models that heavily

weigh the strong correlations among MFCCs could yield higher sensitivity and specificity for *read* speech. These models would benefit from the uniform changes in voice quality that read speech seems to convey in depressed individuals. For spontaneous speech, incorporating a broader feature set – including energy and voicing parameters – appears critical due to the greater variation and subtlety in emotional expression. Detection systems might need to apply more sophisticated algorithms capable of discerning the less pronounced, yet more varied, indicators of depression present in spontaneous communication.

Given these observations, it is recommended that depression detection systems integrate modality-specific features to significantly improve performance. Tailoring algorithms to accommodate the structured nature of read speech or the expressive variability of spontaneous speech can leverage the strengths of each modality, thereby enriching the system's diagnostic capability. By doing so, these systems can become more nuanced and accurate, reflecting the complex ways depression manifests in speech.

4.2. RQ2 — Which machine learning algorithms perform best in detecting depression from speech, and how do they compare to current clinical assessment methods?

To answer RQ2, the performance of the deep learning models on *read* and *spontaneous* speech types were evaluated using the various algorithms developed for the audio (Section 3.3.1) and text (Section 3.3.2) models. This allowed for identifying subtle nuances in each algorithm's detection capability, which may improve the screening and diagnostic procedures used in clinical settings.

4.2.1. Audio models

In addressing Research RQ2 regarding the audio models, we will discuss the results in two main sections. First, we will conduct a comparative analysis of simple and advanced CNN architectures. Second, we will present the comparative analysis of all models developed, which includes CNN architectures, recurrent neural network (RNN) architectures, transformer neural networks, and hybrid architectures.

As shown in Table 5, we present the results of the baseline 1D CNN alongside the ResNet and DenseNet-based CNN models for our comparative analysis. These advanced CNN networks were designed to extract deeper and more nuanced feature information, addressing the need for improved feature representation in diagnosing depression. Each model was evaluated on both tasks, with detailed results provided. Some key observations are presented below.

- **1D CNN:** Achieved the highest Recall (84.48%) in the Reading Task, indicating its effectiveness in identifying positive cases of depressive speech. It outperformed both DenseNet and ResNet models in the Interview Task, achieving an F-score of 81.67%. However, it showed a slightly lower F-score (77.05%) in the Interview Task compared to DenseNet, indicating room for improvement in balancing precision and recall.

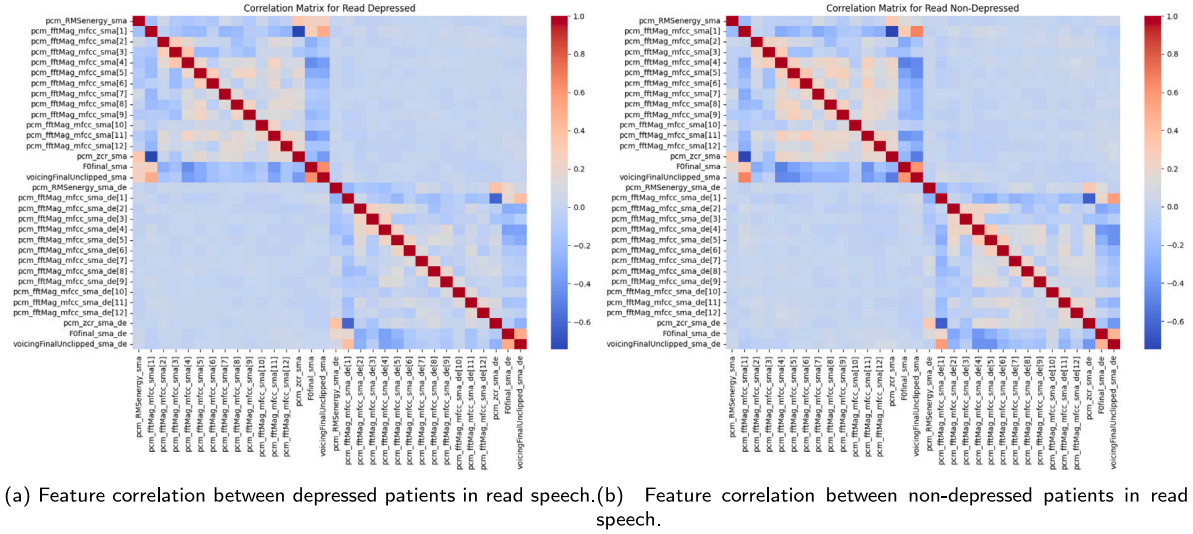


Fig. 10. Feature correlation in read speech.

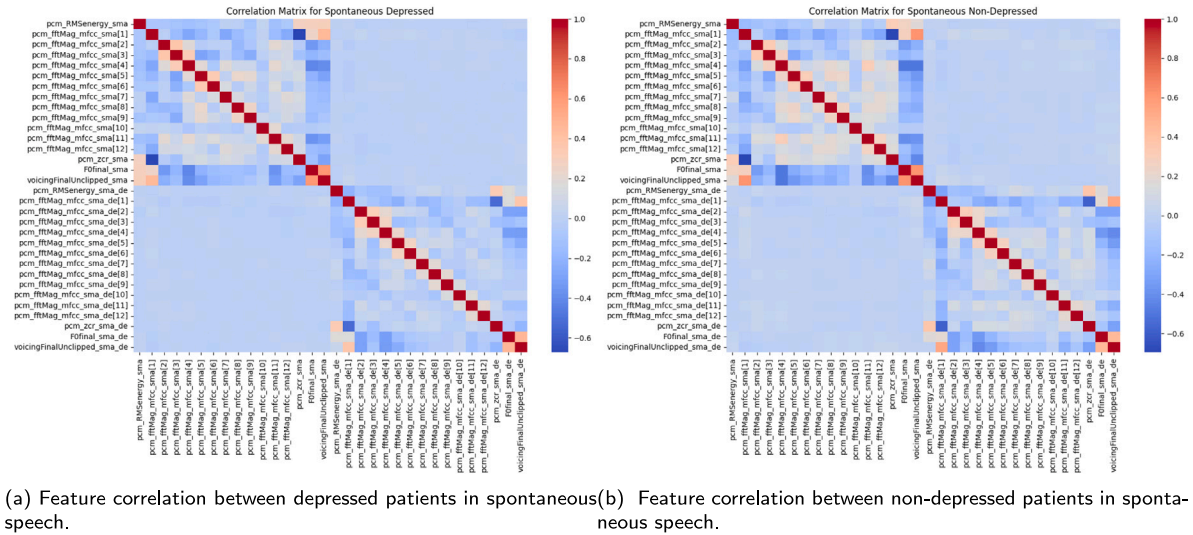


Fig. 11. Feature correlation in spontaneous speech.

- **DenseNet CNN:** Demonstrated strong performance in the Interview Task, achieving the highest F-score of 80.33% and showcasing its ability to handle complex feature relationships. However, it slightly underperformed in the Reading Task compared to 1D CNN and ResNet, suggesting limitations in effectively handling simpler patterns.
- **ResNet CNN:** Consistently performed well across both tasks, achieving the highest Precision (85.19%) in the Interview Task. The average F1-scores of the 1D, DenseNet, and ResNet CNN models are 79.36%, 78.69%, and 79.67%, respectively. This indicates that all three architectures are competitive, with ResNet CNN providing slightly better overall performance. None of the CNN architectures exhibited superior performance in both tasks, indicating that the best choice depends on the specific problem being addressed and the characteristics of the dataset. For the Reading Task, the 1D CNN and ResNet CNN are preferable, while in the Interview Task, DenseNet CNN significantly outperforms both the 1D and ResNet CNNs.

The Hybrid CNN-LSTM model outperforms all other audio models in both the Reading Task (RT) and the Interview Task (IT), as shown

in Table 5. Specifically, for RT, where the structured and consistent speech is conducive to audio analysis, the model achieves an F-score of 84.03%, with an accuracy of 83.04%. This suggests that the model is particularly effective at capturing the less varied acoustic features typical of read speech. In the more spontaneous and variable IT, the model's performance achieves an F-score of 81.97% and an accuracy of 81.03%. Despite the challenges posed by the greater variability in conversational speech, the Hybrid CNN-LSTM model's architecture, which combines CNN's feature extraction capabilities with LSTM's sequential data processing strength, allows it to outperform algorithms for speech-based depression detection.

4.2.2. Text models

As shown in Table 6, the BERT model significantly outperforms other text models in both tasks. During the RT, the robust performance of the model, with an accuracy of 82.84% and an F-score of 85.09%, indicates that it is highly effective even when handling the more controlled linguistic content typical of read speech. This high F-score demonstrates that when the BERT model identifies a case as depressed in the reading context, it is highly likely to be accurate. However, the performance suggests that the restricted nature of read speech

Table 5

Performance of proposed audio models across tasks.

Audio model	Task	Accuracy	Precision	Recall	F-Score
PID CNN	RT	80.36	79.03	84.48	81.67
	IT	75.86	81.03	73.44	77.05
PDenseNet CNN	RT	75.00	73.44	81.03	77.05
	IT	79.31	84.48	76.56	80.33
PResNet CNN	RT	80.36	80.00	82.76	81.36
	IT	77.59	85.19	71.88	77.97
LSTM	RT	79.46	80.63	80.79	79.07
	IT	75.86	80.01	74.00	75.97
Bi-LSTM	RT	81.25	80.67	80.13	80.01
	IT	81.03	84.10	78.56	80.80
GRU	RT	79.46	79.39	79.44	79.48
	IT	66.38	70.95	66.97	67.42
Transformer	RT	76.97	76.67	79.31	77.97
	IT	72.65	74.63	76.92	75.76
HybridCNN-LSTM	RT	83.04	81.97	86.21	84.03
	IT	81.03	86.21	78.13	81.97
HybridCNN-Bi-LSTM	RT	77.68	88.13	74.14	77.48
	IT	77.59	83.93	71.88	75.41

Table 6

Performance of proposed text models across tasks.

Model	Task	Accuracy	Precision	Recall	F-Score
BERT	RT	81.25	87.76	74.14	80.37
	IT	93.10	98.28	89.06	93.44
CNN with BERT embeddings	RT	58.04	63.37	57.99	57.59
	IT	73.28	75.07	73.15	73.66
CNN with TF-IDF features	RT	52.68	57.42	52.65	53.18
	IT	73.28	74.99	73.26	73.22
LSTM with BERT embeddings	RT	58.93	68.68	59.05	58.70
	IT	76.72	77.72	76.70	76.66
LSTM with TF-IDF features	RT	52.68	57.42	52.65	58.70
	IT	68.10	82.69	68.10	67.79

might limit the model's ability to fully utilise its analytical capabilities, compared to more dynamic interactions.

In IT, where individuals may divulge more about their emotional state through varied linguistic cues, the BERT model achieves an accuracy of 93.10% and an F-score of 93.44%. The performance in IT shows the model's strength in textual analysis, enabled by its design optimised for contextual understanding. This allows the BERT model to effectively identify depression markers from the richer linguistic context offered by spontaneous speech. These metrics show that the BERT model is effective in detecting the subtle linguistic indicators of depression often found in spontaneous interactions, making it a suitable tool for evaluating patient speech transcripts or written communication for signs of depression. The better performance in IT also highlights its practical utility in real-world clinical settings where conversational speech is prevalent, illustrating the model's significant advantages in environments that mimic natural human interactions.

4.2.3. Comparison with current clinical assessment methods

When comparing these machine learning models to traditional clinical assessment methods, it is evident that they provide a more effective approach than the average General practitioner's (GP's) unassisted diagnosis, which, according to a 2014 study [63], has a sensitivity of 51%. The Hybrid CNN-LSTM and BERT models outperform this clinical baseline regarding their F-scores, highlighting their capacity to balance precision and recall effectively. This is crucial for enhancing the accuracy of depression screening, suggesting that patients who might otherwise be overlooked in routine clinical evaluations could be more accurately identified with these advanced computational techniques.

Table 7

Comparison of best-performing decision-level fusion technique and feature-level fusion.

Fusion method	Task	Accuracy	Precision	Recall	F-Score
Decision-level	RT	92.86	98.08	87.93	92.73
	IT	94.83	96.77	93.75	95.24
	Both	94.30	97.40	91.80	94.51
Feature-level	RT	57.14	57.81	63.79	60.66
	IT	52.86	53.01	54.00	53.50
	Both	54.96	55.37	58.81	57.01

Table 8

Comparison of unimodal and multimodal performance across the entire dataset.

Model/Fusion method	Accuracy	Precision	Recall	F-Score
Audio model	82.02	84.13	82.10	82.98
Text model	87.28	93.11	81.73	87.02
Decision-level fusion	94.30	97.40	91.80	94.51
Feature-level fusion	54.96	55.37	58.81	57.01

Additional studies have provided the efficiency of different depression assessment instruments such as the patient health questionnaire (PHQ), Children's Depression Inventory (CDI), Geriatric Depression Scale (GDS), and Edinburgh Postnatal Depression Scale (EPDS) used by primary care professionals for most people. Between 78% and 94% of the time, PHQ, including the two-item (PHQ-2) and nine-item (PHQ-9), can assist in the diagnosis of depression [64,65]. Additionally, research indicates that 44% and 76% of the time, a clinical interview using CDI can assist in the diagnosis of depression in children [66]. According to a meta-analysis, GDS can help in the diagnosis of depression in adults with a sensitivity ranging from 74% to 86% [67]. The sensitivity of EPDS is 94%, according to psychometric research [68]. Since PHQ-9 is the universal community screening tool for depression [69,70]. As shown in the evaluation of current screening tools, we observe that analysing read and spontaneous speech is not part of the assessment methods in primary care. Primary care providers have often faced the difficult challenge of identifying the optimal screening approach for depression diagnosis [71], and we believe that the integration of *read* and *spontaneous* speech analysis as described in our approach offers an enhancement for the screening and diagnosis of depression.

4.3. RQ3- which method of multimodal fusion demonstrates the highest performance in detecting depression from speech, and how do they compare to current clinical assessment methods?

To answer RQ3, the performance of early (feature-level) and late (decision-level) fusion techniques was evaluated as detailed below.

4.3.1. Feature-level Fusion (FLF)

FLF was performed using the best-performing model from Table 5 (Hybrid CNN-LSTM), alongside BERT embeddings for text features, which were shown to produce the best results in Table 6. However, Tables 7 and 8 show that FLF led to a significantly worse performance across all metrics than single-mode analysis or other types of fusion. This suggests that the combined feature set may have been too complex or mismatched for the model to process effectively, leading to decreased overall effectiveness in depression detection.

4.3.2. Decision-Level Fusion (DLF)

DLF resulted in a better performance than single mode analysis, as shown in Table 8. Table 9 indicates that the weighted average method performs best across all metrics among the four decision-level fusion techniques evaluated. This suggests that incorporating different decision weights, tailored to the predictive strengths of each model, results in a more accurate and reliable depression detection system.

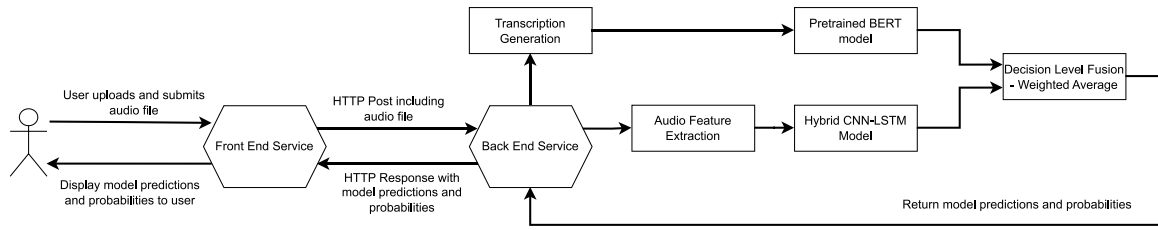


Fig. 12. Proposed deployment of the multimodal depression detection.

Table 9

Performance of decision-level fusion techniques across the entire dataset.

Fusion method	Accuracy	Precision	Recall	F-Score
Weighted average	94.30	97.40	91.80	94.51
Bayesian	93.86	95.00	93.44	94.21
Logistic regression	93.86	96.55	91.81	94.11
Gated multimodal unit	88.57	86.67	86.67	86.67

4.3.3. Comparison of decision-level fusion and feature-level fusion

Table 7 directly compares the best DLF technique — weighted average, and FLF. DLF consistently outperforms FLF across all metrics, indicating that DLF, which combines the outputs of individual models at a later stage, is more effective at classifying depressive cases than FLF, which combines data at the input level before processing.

4.3.4. Comparison to clinical assessment methods

The weighted average decision-level fusion method, by demonstrating a better F-score over the current methods used for diagnosing depression mentioned in Section 4.2, suggests potential as a supplementary tool that could enhance the accuracy of GPs' diagnoses. Fig. 12 illustrates how it could be integrated into clinical practice, providing a simplified approach for GPs to utilise multimodal data in their diagnostic process. This could be incorporated into the healthcare system for the initial depression screening process. GPs could upload audio recordings of the patient and use the returned probabilities as a second opinion. Integrating such advanced computational tools could enhance primary care settings' screening and diagnosis process, aiding GPs by providing an objective, quantifiable second opinion. This alignment could improve patient outcomes by reducing the likelihood of untreated depression, a treatable condition with significant personal, healthcare system, and societal implications [2]. However, integrating such tools in primary care settings would require consideration of the privacy, security and ethical implications of processing patient audio data within the electronic health records (EHR) systems, GP training and a cost-effectiveness analysis of the sensitivity and specificity of the system for depression screening, diagnosis and treatment.

4.3.5. Proposed workflow of the integration in primary health care

The proposed integration of the automatic depression screening tool into a hospital or primary care system aims to enhance the efficiency and effectiveness of depression diagnosis. Fig. 13 is a detailed description of the operational workflow, complementing the diagram of the deployed solution in Fig. 12, that outlines the patient journey through the primary care system with the addition of the proposed solution.

When a depression screening is initiated at the automatic screening phase, audio recordings of the patient speaking are uploaded, automatically transcribed and processed using the deployed solution in Fig. 12 to assess the likelihood of depression. The screening output can include a risk score based on probabilities or a recommendation for further evaluation, which the clinician reviews to help make an informed decision regarding the patient's mental health based on the audio data. The tool's outputs and decisions made by the clinician will be recorded in the patient case notes within the EHR to ensure continuity of care and engagement with the patient.

Table 10

Previous unimodal approaches involving the same participants.

Study	Accuracy	Precision	Recall	F-Score
[3]	77.0	74.0	80.0	77.0
[4]	84.5	84.5	84.6	84.5
[5]	67.6	71.7	72.2	72.3
[6]	84.4	84.5	85.6	83.7
[7]	88.0	88.1	89.2	87.7
Ours	94.3	97.4	91.8	94.5

The operational workflow streamlines the process of depression screening and diagnoses in primary care settings. Because the system is automated, there is a reduction in the time between symptom identification and diagnosis system as the data analysis and result synthesis are done in real-time. This could lead to better patient outcomes and satisfaction. Using on-the-fly advanced machine learning algorithms enhances diagnostic accuracy, reducing human error in screening and diagnosing depression. The proposed integration also serves as a triage mechanism, as clinicians can initiate screening without specialised training in mental diagnostics, and depending on the results the clinician may refer the patient to a mental health specialist, prescribe treatment, or schedule follow-up visits.

4.4. RQ4 -How does integrating linguistic and paralinguistic features enhance the accuracy and reliability of depression detection compared to single-mode speech analysis?

The proposed multimodal approach, which integrates both linguistic and paralinguistic features for the detection of depression from speech, significantly outperforms all single-mode models developed in this research, as shown in Table 8, which compares the performance of the best performing unimodal models with the multimodal fusion techniques performed. The DLF method improves upon the metrics of each model, suggesting that the fusion at the decision level effectively leverages the strengths of both audio and text modalities. In contrast, FLF results in worse outcomes than the two best-performing individual models. This might indicate that early fusion could dilute or misalign critical features when prematurely combined, leading to suboptimal performance.

The multimodal approach also outperforms all previous unimodal (audio-only) methods that utilise the *Androids Corpus* [6] benchmark for speech-based depression detection, as shown in Table 10. A key advantage of the multimodal methodology is the inclusion of both read and spontaneous speech samples. For instance, the earlier studies referenced in Table 10 and Table 11 all relied on only one type of speech — either *read* or *spontaneous* — which may not encompass the natural variations present when both speech types are included. In contrast, the current approach, which consists of both read and spontaneous speech, captures a broader range of emotional expressions and speech patterns, enhancing the robustness and applicability of the findings.

Due to variations in experimental protocols and the type of speech data used, establishing a fully rigorous comparison with previous research is challenging. Nevertheless, this study reports the highest accuracy, precision, recall, and F-scores as shown in Table 10, highlighting

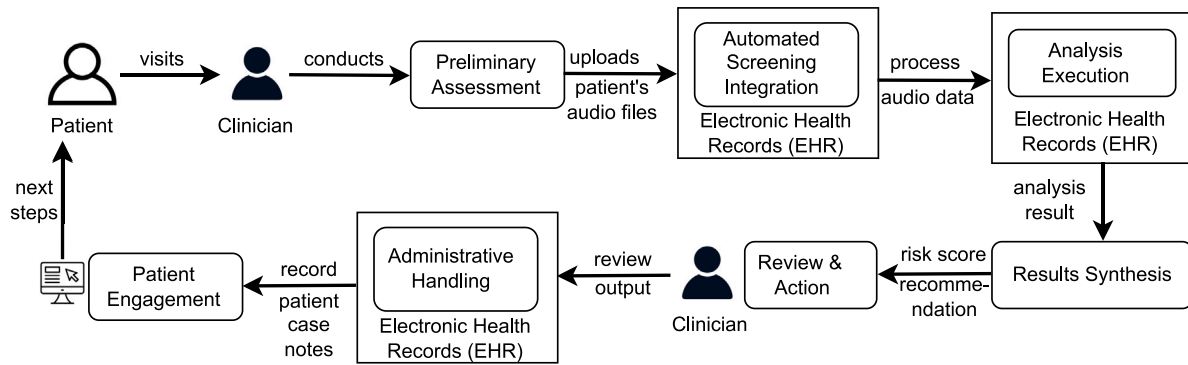


Fig. 13. Operational workflow of automatic depression detection in primary care settings.

Table 11

Previous multimodal approaches involving the same participants.

Study	Accuracy	Precision	Recall	F-Score
[72]	83.5	95.0	70.3	80.5
[73]	83.0	95.2	69.0	80.0
[26]	84.7	95.2	72.4	82.3
Ours	94.3	94.7	91.8	94.5

the benefits of multimodal fusion and the inclusion of diverse speech types. Specifically, it achieves a detection accuracy of 94.3%, exceeding the highest unimodal accuracy of 88.0%. The F-score, which combines precision and recall into a single measure, is 94.5% under the multimodal approach, significantly higher than the unimodal peak of 87.7%. These results underscore the effectiveness of combining linguistic information from the spoken content with audio-based paralinguistic cues, and the added value of analysing both read and spontaneous speech. This evidence strongly supports the utility of a multimodal framework in developing more advanced and sensitive tools for mental health screening and diagnostics.

4.5. RQ5 — How does the proposed multimodal depression detection technique improve upon current methodologies in the field of affective computing?

As shown in Table 11, the accuracy in the proposed multimodal approach increased to 94.3%, up from a range of 83.0% to 84.7% in earlier studies, while the F-score increased notably to 94.5%, demonstrating a well-balanced approach between precision and recall. Integrating read and spontaneous speech, the proposed approach provides a more holistic view of speech as a diagnostic tool. This integration allows the model to analyse and learn from the contrasts and similarities between these two types of speech. For example, while read speech might offer a baseline of how an individual articulates words in a controlled environment, spontaneous speech can provide additional context on how their speech patterns change when they are less guarded or more emotionally expressive.

By focusing on these differences, the research achieves a refinement of the sensitivity of the detection algorithms to subtle indicators of depression that may be more prevalent in spontaneous interactions. This could include detecting patterns of speech that correlate with depressive symptoms, such as monotonous tone, slowed speech rates, or reduced phonetic variability, which might be less apparent or altogether absent in read speech. Therefore, the emphasis on exploring the interplay between these speech modalities enriches the dataset and enhances the algorithm's ability to generalise across different speaking conditions. This not only improves the accuracy and robustness of depression detection but also offers insights into the complex ways depression can affect communication, contributing valuable knowledge that could influence both the diagnosis and treatment of depression in clinical settings.

5. Limitations

Despite the significant outcomes of the multimodal framework, the research is subject to several limitations that merit consideration for future investigations. A primary concern is the reliance on automated transcription services without subsequent verification by native language experts. This methodological oversight introduces a potential vector for error, as even advanced transcription algorithms such as OpenAI's Whisper and Google's Cloud Speech API may not fully capture the nuanced linguistic features specific to Italian. Misinterpretations or omissions during transcription can lead to inaccuracies in the training and validation phases of the model development, potentially compromising the model's diagnostic precision. Additionally, the research did not extend the feature correlation analysis to textual data due to potential inaccuracies in the automated transcripts, which might have limited the depth of findings related to the textual aspects of speech. The research explored several decision-level fusion techniques to enhance the performance of the predictive model, however, not all available fusion methods were experimentally evaluated. Techniques such as 'Concat', which involves concatenating unimodal scores, 'Concat+MultiTask', applying a Multi-Task loss, 'Moddrop', designed to manage modality-specific dropout [74], and 'CentralNet' which uses multi-task learning to regularise the modality-specific neural networks [75], was not included in the analysis. These untested methods represent additional approaches that could potentially influence the outcomes and effectiveness of multimodal fusion.

Other multimodal fusion techniques based on neural networks, such as The Sketch [51], which includes both classical and binarized versions, and the early fusion models that concatenate the embeddings of each modality at the input level were not considered. Furthermore, the research did not cover hybrid fusion approaches, which combine the benefits of early and late fusion methods as a multimodal fusion strategy when evaluating audio and textual data for depression identification. The study's focus on the Italian language inherits the limitation associated with low-resource scenarios such as non-English settings affected by data scarcity conditions, even though it offers insightful results about a linguistic context that is not as widely explored. For instance, there are 228 recordings in our corpus. 75 Italian cardiology reports were used in [15] to develop a clinical information retrieval system. *PsyNIT*, the first Italian Named Entity Recognition dataset in neuropsychiatry was developed using 100 medical reports [76]. The fact that only 10,000 Arabic tweets were included in the published Arabic depression corpus¹³ in [16] impacts how broadly the findings may be applied to the Arabic-speaking community. While our solution demonstrates the suitability of the approach in a scenario with limited data availability and compatibility with less-resourced languages like

¹³ <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/YHMYEQ>.

Italian, the relationship between the limited dataset and the observations requires further investigation. Moreover, utilising a clinically controlled dataset, while beneficial for ensuring data consistency and reliability, may not encompass the broader spectrum of speech variability encountered in more naturalistic settings. This limitation could affect the model's performance in real-world applications, where speech patterns may be more varied and less predictable. Future research will incorporate datasets from various languages and cultural backgrounds.

6. Conclusion

This research has contributed to knowledge in automatic depression detection through the innovative integration of Long Short-Term Memory Networks, Convolutional Neural Networks, and Bidirectional Encoder Representations from Transformers within a multimodal analytical framework. By utilising a robust benchmark dataset for speech-based depression detection, comprising both read and spontaneous speech samples in Italian, this study achieved a diagnostic accuracy of 94.30% and an F-score of 94.51%. These results affirm the proposed model's effectiveness and represent an advancement over existing benchmarks in computational paralinguistics and mental health diagnostics.

One of the strengths of this research lies in its detailed analysis of the feature correlation between read and spontaneous speech in depressed individuals using audio files. This nuanced approach provided valuable insights into how different speech modalities reveal unique, depression-related acoustic patterns. By isolating and comparing these modalities, the study highlighted distinct variations in spectral patterns and vocal features, which could significantly enhance the accuracy and specificity of depression detection models. Future work will focus on the following areas: the verification of transcriptions to improve the accuracy of the textual models; experimentation with other multimodal fusion techniques; a diverse dataset across multiple languages and cultural contexts to facilitate the development of more universally applicable diagnostic tools; and the inclusion of naturalistic speech data outside of clinical settings to enhance the ecological validity of the research, making the findings more applicable to everyday clinical practices.

CRediT authorship contribution statement

Klara Daly: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation. **Oluwafemi Olukoya:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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